

Sara Miller

Phone : (416)-475-8447

Email : saramiller1797@gmail.com

LinkedIn: www.linkedin.com/in/sara-miller17

EDUCATION

BESc. Mechatronic Systems Engineering *University of Western Ontario, London Ontario* 2022 - Expected May2027

- Granted distinction of Dean's List for high academic achievement

WORK EXPERIENCE

Mechanical Design Intern, Honda of Canada Mfg. May 2025 - Aug 2026

- Mechanical design lead for multiple automation projects, owning design for robotic tooling, manufacturing fixtures, jigs, and production equipment from concept development through fabrication, installation, and validation.
- Created and maintained CATIA V5 part models, large assemblies, detailed drawings, and revision-controlled design packages used by suppliers, manufacturing, maintenance, and installation teams.
- Applied FEA, rigidity calculations, robot reach studies, payload calculations, and interference checks to validate designs and reduce technical risk before implementation.

Mechatronic Systems Engineering Co-op Student, Honda of Canada Mfg. May 2024 - Aug 2024

- Developed and delivered a course on pneumatic systems: circuit diagrams, and troubleshooting techniques
- Gained foundational experience in Mitsubishi PLC programming, supporting small-scale automation updates and troubleshooting logic-based control systems.
- Developed foundational experience with Cognex machine-vision and vision-guidance systems integrated with robotic automation.

High Ropes Team Lead, Blue Mountain Resort May 2023 - Aug 2023

- Led a team of 20 high ropes course guides, fostering a collaborative and safety-focused environment
- Acted as the primary point of contact for guest inquiries and concerns, consistently resolving issues to maintain a high level of customer satisfaction
- Maintained up-to-date knowledge of industry safety standards and ensured compliance with all relevant regulations and protocols

Sales Associate, SVP Sports 2020 - 2024

- Responsible for end of day cash tallying and balance confirmation
- Managed inventory, including daily receiving and shipping of product orders, trained new employees

EXTRACURRICULARS

Head of Technical Communications, Western Engineering Concrete Canoe Association Sept 2025 - Aug 2026

- Led the creation and management of technical documentation across multiple sub-teams, ensuring consistent communication of project workflows and producing detailed technical reports.

Big Sister Mentor, Western Woman in Engineering Club Sept 2023 - May 2025

- Mentored incoming first year engineering student providing academic guidance fostering supportive community.

Orientation Week Leader, Engineering SOPH Feb 2026 - Present

- Mentored first-year engineering students, fostering an inclusive and supportive community through transition programming, group activities, and peer guidance.

SKILLS

CAD/Design: CATIA V5, DELMIA V5, SolidWorks, AutoCAD Electrical

Programming/Controls: Java, C, C++, MATLAB/Simulink, Mitsubishi PLC

Engineering Methods & Analysis: GD&T, FEA, DFMA, Pneumatic Circuit Design, Payload Calculations, Rigidity Calculations